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Adsorption of hydrogen on the Pt(111) surface from low-energy recoil scattering

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Abstract

We have shown that low-energy recoil scattering (LERS) can be used to detect hydrogen on a Pt(111) surface at 150 K and that this method provides a direct probe for locating the positions of adsorbed hydrogen on a surface. It was found: (1) the initial sticking coefficient of a H atom is $s_0(\theta) \approx 0.1$; (2) from the isothermal annealing, the activation energy of H atoms adsorbed on the surface is evaluated to be 0.3 eV; and (3) the H atoms occupy fcc-adsorption sites on the Pt(111) terraces at a height of approximately 1.2 Å above the first layer of Pt atoms. © 1997 Elsevier Science B.V.

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1. Introduction

The interaction of hydrogen with solid surfaces is of wide scientific and technological interest, and many techniques have been used to study hydrogen on surfaces.

For more than 20 years, the most powerful quantitative technique for hydrogen analysis in solids has been resonance nuclear reaction analysis (RNRA) which used $^1H(^{15}N, \alpha\gamma)^{12}C$ at 6.385 MeV established by Lanford et al. [1,2]. The second popular technique is elastic recoil detection analysis (ERDA) in which MeV ions bombard a sample, and through elastic collisions cause hydrogen ions to recoil out of the sample [3,4]. For most problems where the need is to determine the amount of hydrogen, these techniques are generally agreed to have the best analytic capabilities.

In addition, the determination of the surface structure of the adsorbate–metal system is a topic of continuing interest in surface science. Studies of hydrogen adsorption on metal surfaces have been carried out by low-energy electron diffraction (LEED) [5], work function variation [5], electron energy loss spectroscopy (EELS) [6], thermal programmed desorption [5,7,8], photoemission and helium diffraction [9]. These techniques provide detailed information on the interaction of hydrogen with surfaces. However, it is still difficult to probe the location of the adsorbed hydrogen

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on surfaces using these surface structural analytic tools. In contrast, low-energy recoil scattering (LERS) techniques offer the possibility of directly observing and locating surface hydrogen by analyzing the angular anisotropy of recoiled hydrogen. The concept of shadowing and blocking based on simple classic mechanics is very effective for surface structure analysis [10–14]. In a recoil process, the target atom is kicked out of the surface in a binary encounter at a relatively high energy.

Platinum is one of the most important materials as a catalyst in hydrogenous reactions. The interaction of hydrogen with platinum surface has been extensively studied [5–9], but the process and properties of hydrogen adsorption on platinum are still not completely known nor understood. For example, LEED measurements are not applicable detecting the formation of any ordered phase of hydrogen on Pt(111).

In what follows, we report the results found from experiments designed to detect hydrogen on a Pt(111) metal surface by LERS using 1 keV–Ne^+. In particular, we discuss the hydrogen adsorption, desorption processes, and information on the position of hydrogen on Pt(111) at 150 K coupled with trajectory simulations. Calculations were performed using the Thomas–Fermi–Moliére (TFM) potential [15] with Frisov's screening length a_F [16], and using procedures described in other papers [17–19]. In addition, the experimental value of the incident critical angle is defined as the angle where the intensity curve attains the 70% maximum [18]. In this paper, symbols of "$\alpha_{c,sh}$" and "$\alpha_{c,bl}$" represent the critical incident angle versus α scan determined by shadow cones and blocking cones, respectively.

2. Experimental procedures

The experiments presented here were performed in an ultra-high vacuum system with LEED, AES and LERS facilities as shown in Fig. 1. The base pressure during the experiments was maintained below 3×10^{-8} Pa. A 12 mm-diameter Pt(111) single crystal was mounted on a standard *xyz*-manipulator after mechanical and electrochemical polishing. The crystal was cleaned in situ by

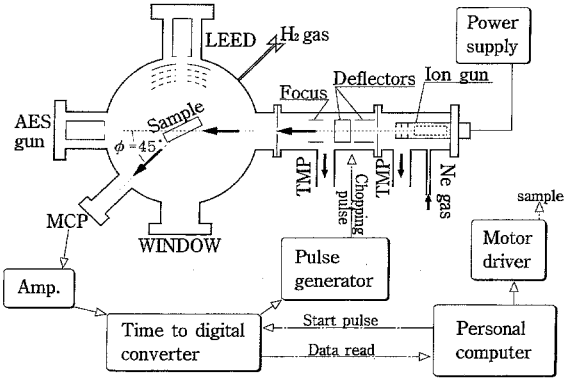


Fig. 1. Spectrometer system designed for LERS analysis along with conventional LEED and AES analysis.

repeated cycles of 500 eV–Ar^+ bombardment followed by annealing at 600°C to remove the surface damage. After this treatment no carbon, oxygen or sulfur surface impurities were detectable by AES and LEIS. Sharp (1 × 1) LEED patterns were thus obtained. Hydrogen adsorption onto the sample was made by introducing H_2 gas into the chamber at 150 K. The amount of H_2 exposure was accurately controlled by an automatic leak valve at the hydrogen gas partial pressure of 1.33×10^{-4} Pa (1.00×10^{-6} torr).

The LERS flight time spectra were taken by chopping the primary 1 keV–Ne^+ beam with a pulse width of about 100 ns and the forward scattered particles, after a free flight through a drift tube of 30 cm, were recorded by a micro channel plate (MCP). As indicated in Fig. 2, the detection system was located at 45° with respect to the incident beam. The recoiled hydrogen were

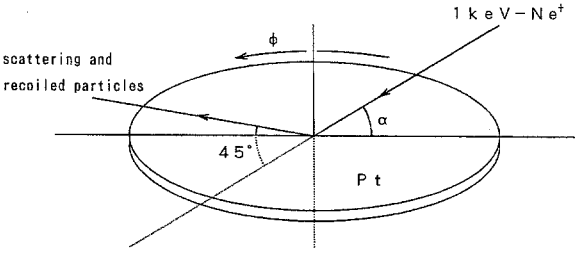


Fig. 2. Details of geometrical arrangements taken for the investigation of a hydrogen adsorbed Pt(111). The symbols α and ϕ show polar and azimuthal angles, respectively. A detection angle is fixed at 45° with respect to the incident beam direction.

detected as a function of polar angle α, and azimuthal angle ϕ between the Ne^+ ion beam incident direction and the crystal axes. A typical time-of-flight spectrum of recoiled hydrogen is shown in Fig. 3. The peaks observed at around 9.3 μs and at around 9.6 μs correspond to surface recoil (SR) and direct recoil (DR) hydrogen due to the different recoil processes, respectively. The SR feature is very sensitive to the position of adsorbed hydrogen atoms with respect to the nearest neighboring Pt atoms. Another signal at around 10.7 μs comes from the primary Ne beam scattered by the Pt atoms. Polar angle, α (impact angle) scans were performed from 40° through 5° in 1° decrements with an average beam current of ~ 10 pA. In addition, azimuthal angle, ϕ, scans were performed between $\pm 66°$ in 3° increments. Time requirements for a complete scan was typically 30 min giving a total dose of $\sim 5 \times 10^{11}$ ions cm^{-2}, which is low enough to avoid radiation damage due to the primary beam.

3. Results and discussion

3.1. Hydrogen adsorption and desorption processes

Information on the kinetics of adsorption were obtained from evaluating the adsorbed amounts of hydrogen as a function of H_2 exposure. The increase of the H coverage as a function of H_2 exposure at 150 K is plotted in Fig. 4. The axis labeled "H intensity" is the sum of the DR and SR events in the time-of-flight spectrum. The insert of Fig. 4 shows the variation of hydrogen intensity as a function of time after stopping the hydrogen exposure to the Pt(111) sample. An exposure of about 50 L (Langumuirs) H_2 at 150 K was sufficient to saturate the surface with adsorbed hydrogen. Modeling the data in Fig. 4 shows an initial sticking coefficient, $s_0(\theta) \approx 0.1$ is in agreement with previous work [5]. The simplest phenomenological model for the rates of hydrogen gas adsorption and desorption may be described by a differential equation of the form

$$dN/dt = -k_{de}N + k_{ad}(N_{sa} - N), \quad (1)$$

where k_{de}, k_{ad} and N_{sa} are the desorption, adsorption coefficients and sa the saturated number of H atoms, respectively. The left side of Eq. (1) shows the rate of increment of the H atoms. At the condition of $N=0$ at $t=0$ s, Eq. (1) is easily solved as

$$N = \{k_{ad}N_{sa}/(k_{de} + k_{ad})\} \times [1 - \exp\{-(k_{de} + k_{ad})t\}]. \quad (2)$$

Since LEIS and LERS lack the ability of quantitative analysis, the value of saturated H_2 coverage

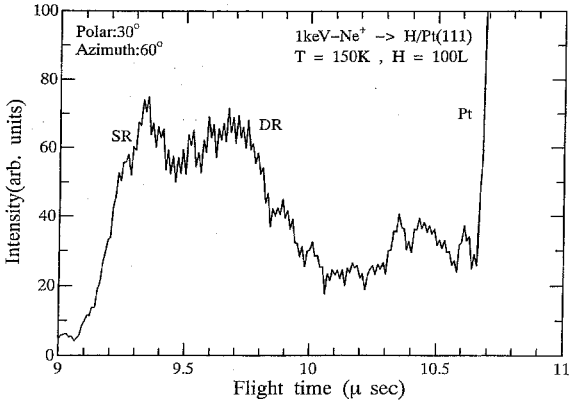


Fig. 3. Typical energy spectra of recoiled hydrogen as a function of the flight time. The beam incidence is $\alpha = 30°$ and $\phi = 60°$ [$2\bar{1}\bar{1}$] to the sample surface. The first peak marked SR and the second peak marked DR show surface and direct recoil hydrogen, respectively.

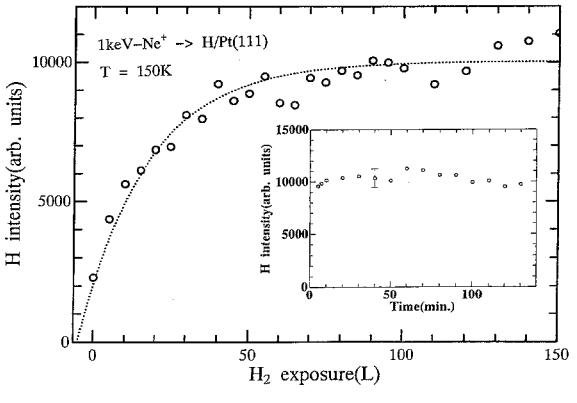


Fig. 4. Variation of the coverage with hydrogen exposure at 150 K for a Pt(111) surface. The dotted line shows the calculated values by using Eq. (2). The insert of the figure shows the H intensity versus time after stopping the hydrogen exposure to the sample.

at 150 K, as $N_{sa}=1.5\times10^{15}$ atoms cm^{-2} was taken from the saturation D_2 coverage at 180 K as determined by high-energy elastic recoil detection analysis [20]. This value corresponds to 1.0 ML (monolayer) of the ideal Pt density in a (111) plane of the bulk crystal. This value is reasonable in our case because of the well-ordered (1×1) LEED pattern seen with the saturated hydrogen adsorption at 150 K. In addition, in this case k_{de} can be neglected since no hydrogen desorption can be seen, as shown in the insert of the Fig. 4. This result is similar to previous work observed by thermal desorption technique [5]. The values of k_{ad} evaluated by fitting to the data in Fig. 4 is 5.2×10^{-2} s^{-1}. Fig. 5 shows the variation of hydrogen intensity as a function of annealing temperature for the stepwise heating of 3 min at every 10 K. Assuming hydrogen adsorption follows an Ahrennius relation, then substituting the measured isosteric heat of adsorption into the second-order desorption kinetics equation [21,22] becomes

$$d\theta/dt=-v\theta_2\exp(-E_d/kT), \quad (3)$$

where θ is the coverage of hydrogen atoms (N/N_{sa}), v is the frequency factor, E_d is the activation energy of desorption and k is the Boltzmann constant. Fitting Eq. (3) to data in Fig. 5 using $v=10^{13}$ s^{-1} gives $E_d=0.3$ eV. This is also in agreement with the value of 0.27 eV in a previous report [23]. These experimental results show that LERS is a powerful technique for studying adsorbed hydrogen on a surface. The unique features are direct monitoring of surface hydrogen and the extreme sensitivity to outer most atomic layers.

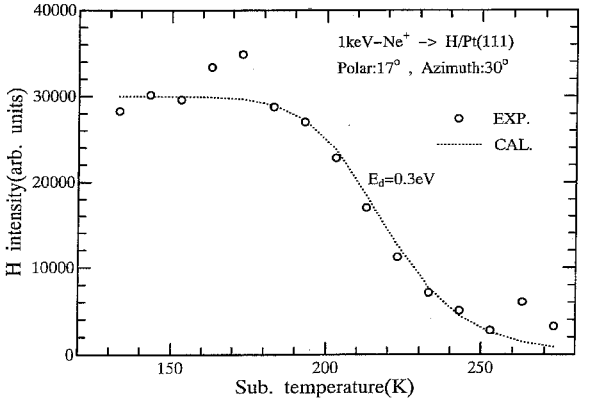


Fig. 5. Variation of the hydrogen coverage during isothermal annealing. The beam incidence is the $[10\bar{1}]$ azimuth of the sample surface. The dotted line is obtained from Eq. (3) with an activation energy of 0.3 eV.

3.2. *Information on the site occupancy of hydrogen on Pt(111)*

Variation in the hydrogen scattering signal with crystal alignment provides direct information about the hydrogen location on the Pt(111) surface. While the SR data should provide information of the specific position of adsorbed hydrogen atoms with respect to the nearest-neighbor Pt atoms, the SR process is complicated making unambiguous interpretation difficult. Hence, we will deduce H/Pt(111) structural information from analysis of the DR data as follows. Fig. 6 shows the variation of the intensity of DR hydrogen as a function of ϕ along the polars $\alpha=18$, 25 and 30° of the Ne^+ ion incident direction. The intensity drops observed at $\phi=0°([11\bar{2}])$, $\phi=60°([2\bar{1}\bar{1}])$ and $\phi=-60°([\bar{1}2\bar{1}])$ show the shadowing effects for $\alpha=18°$, 25° and blocking effects for 30°, respectively. In addition, the intensity peaks around $\phi=\pm40°$ are due to the focusing effect. The magnitude of the intensity at $\phi=0°$ drops off more deeply than that at $\phi=60°$. These profiles clearly show a periodicity of 120° for the azimuthal angle (a

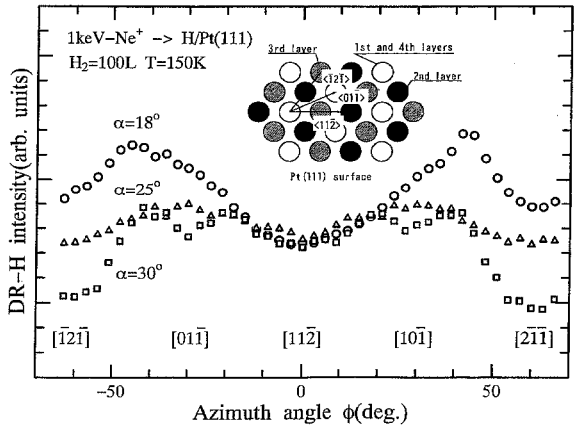


Fig. 6. DR hydrogen intensity profiles as a function of the azimuthal (ϕ) angles along $\alpha=18$, 25 and 30° of the incident beam.

threefold symmetry). They suggest two cases as the adsorption site: (1) the fcc hollow site with third layer Pt atom directly below; and (2) the hcp hollow site with a second Pt atom directly below. Any occupancy of the bridge site or the on-top site can be excluded since a periodicity of hydrogen intensity does not show the twofold symmetry nor the sixfold symmetry but the threefold symmetry. According to our trajectory simulations, hydrogen signals coming from the first layer are only detectable in our experimental geometry. Other hydrogen signals from the second layer or deeper layers cannot be detected because of the shadowing and blocking effects. We therefore do not consider any affections below the second layers of Pt atoms in the following discussion.

Fig. 7 illustrates how the shadowing critical angle can depend on azimuths of the incident direction. In Figs. 7a and b, hydrogen occupies fcc-adsorption sites on the Pt(111) surface. The

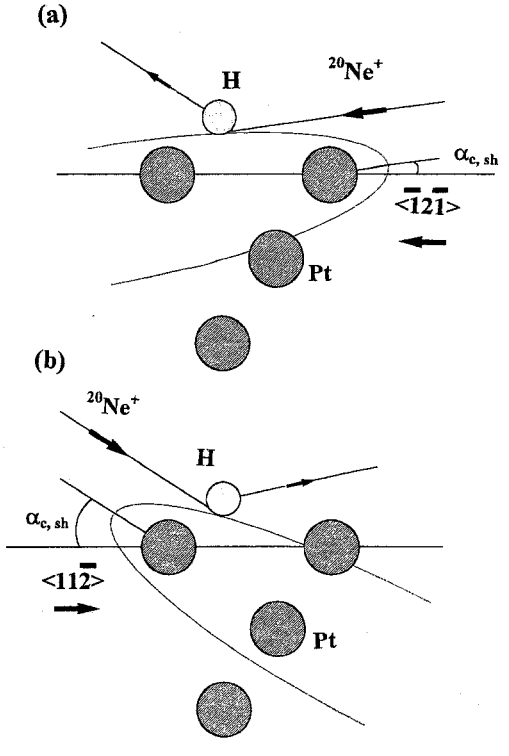


Fig. 7. Schematic diagram showing the $\alpha_{c,sh}$ in both directions of: (a) $\langle\bar{1}2\bar{1}\rangle$; and (b) $\langle11\bar{2}\rangle$. Here, the H atoms occupy the fcc-adsorption sites.

edges of the Pt shadow cones cross directly over the hydrogen atom when oriented at $\alpha_{c,sh}$. One can see that the $\alpha_{c,sh}$ along the $[11\bar{2}]$ azimuth is larger than that along the $[\bar{1}2\bar{1}]$ azimuth.

Fig. 8 shows the variation of the H intensity as a function of the polar angle α obtained from the H/Pt(111) sample along the $[\bar{1}2\bar{1}]$ and $[11\bar{2}]$ azimuths. Strictly speaking, the shape of the blocking cone of the first layer Pt for the scattered Ne beam is slightly larger than that of the shadow cone of Pt for primary Ne beam. Because the Ne beam scattering energy of 985 eV at 45° is smaller than the primary energy of 1 keV. However, assuming both cones have almost the same shape, the experimental data in the region below and above 22.5° (half of detection angle of 45°) are associated with shadowing and blocking effects, respectively. In addition, as shown the angle regions illustrated by shaded areas in Fig. 8, one can see the angle region between $\alpha_{c,sh}$ and $\alpha_{c,bl}$ shifts left (toward the lower angle) in the $[\bar{1}2\bar{1}]$ azimuth and another angle region between $\alpha_{c,sh}$ and $\alpha_{c,bl}$ shifts right (toward the higher angle) in the $[11\bar{2}]$ azimuth.

On the other hand, Fig. 9 shows the calculated values of the H height as a function of the H critical angle α based on the trajectory simulations. (In this figure, α also corresponds to the polar angle.) As shown by arrows, independent of the

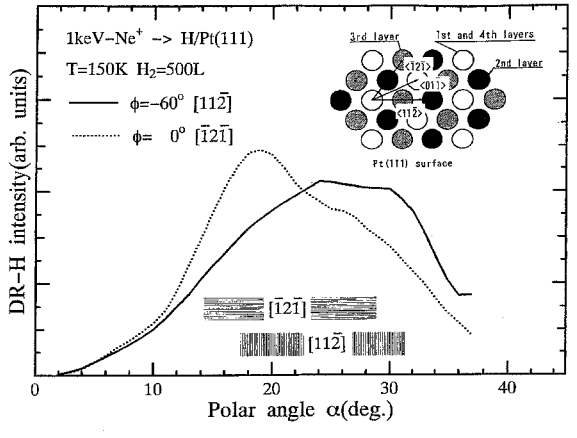


Fig. 8. DR hydrogen intensity profiles as a function of the polar angle obtained from H/Pt(111) samples. The angle regions below and over 22.5° are associated with the shadowing and blocking effects, respectively. The shaded areas show the angle regions between $\alpha_{c,sh}$ and $\alpha_{c,bl}$ in the $[\bar{1}2\bar{1}]$ and $[11\bar{2}]$ azimuths.

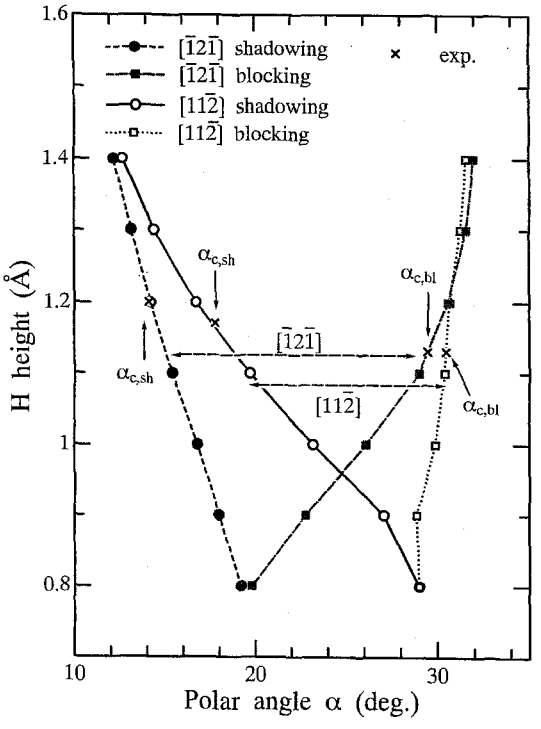


Fig. 9. The calculated values of the H height as a function of the H critical angle α (corresponding to polar angle). The regions between the arrows show the angle between $\alpha_{c,sh}$ and $\alpha_{c,bl}$. The solid and dashed lines are drawn as a guide to the eyes. The "×" symbol represents the experimental values of the $\alpha_{c,sh}$ and $\alpha_{c,bl}$ in the [1̄21̄] and [112̄] azimuths.

H height the angle regions between $\alpha_{c,sh}$ and $\alpha_{c,bl}$ shifts left in the [1̄21̄] azimuth and right in the [112̄] azimuth, respectively. In other words, the calculated value of $\alpha_{c,sh}$ along the [112̄] azimuths is larger than that along the [1̄21̄] azimuths. These trends are in agreement with the trend shown in Fig. 7 and observed from the experimental data shown in Fig. 8. This indicates that the H atoms

Table 1
Observed and calculated critical shadowing and blocking angles. The calculated angles are based on trajectory simulations

		Experimental α_c	Calculated α_c
[1̄21̄]	Shadowing	14.1°	14.5°
[1̄21̄]	Blocking	29.5°	30.0°
[112̄]	Shadowing	17.8°	17.5°
[112̄]	Blocking	30.5°	30.3°

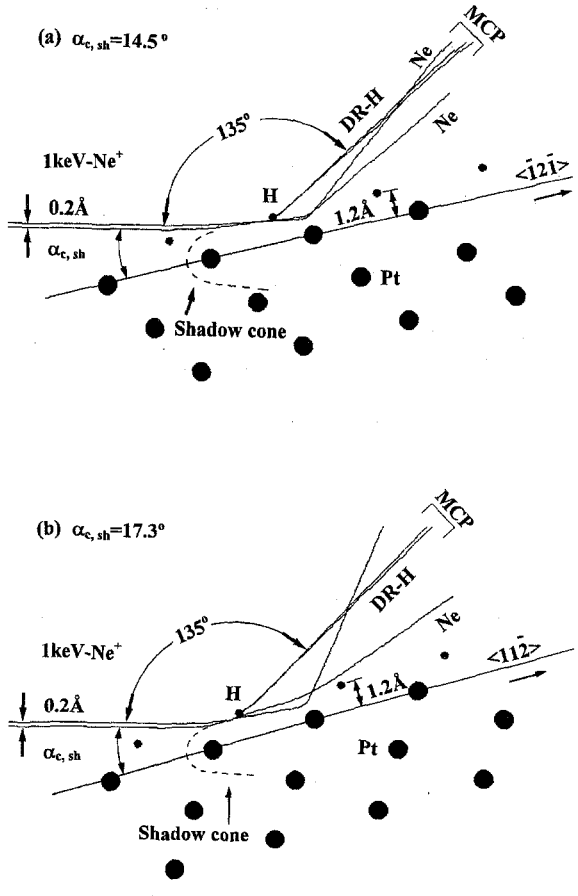


Fig. 10. Calculated trajectories for 1 keV–Ne^+ scattering and DR-H at: (a) $\alpha_{c,sh}=14.5°$ ($\langle 1\bar{2}1\bar{}\rangle$); and (b) $\alpha_{c,sh}=17.3°$ ($\langle 11\bar{2}\rangle$). The vertical spacing of H atoms is 1.2 Å from the first layer Pt atoms. The detectors of MCP are located at the angle of 135° (corresponding to the detection angle of 45°) with respect to the incident beams.

occupy preferably the fcc-adsorption sites. Moreover, the value of H height can be estimated from Fig. 9 by plotting the experimental values of both $\alpha_{c,sh}$ and $\alpha_{c,bl}$ in two azimuths on the calculated data. The H height of 1.16 ± 0.03 Å (≈ 1.2 Å) can be taken as the common value to the observed H critical angles, as summarized in Table 1.

In contrast, we check the possibility of the H atoms occupying the hcp-adsorption sites. If the H atoms took the hcp-adsorption sites, the observed values of $\alpha_{c,sh}$ should be lower than that of the fcc-adsorption sites along the [112̄] azimuths. In other direction [1̄21̄], the value of $\alpha_{c,sh}$

is higher than that of fcc-adsorption sites. These trends are totally the reverse of the experimental data. It indicates that the possibility of the hcp-adsorption sites is excluded.

Fig. 10 shows the trajectory simulations for $\alpha_{c,sh}$ at 14.5° ($\langle\bar{1}2\bar{1}\rangle$) and 17.3° ($\langle 11\bar{2}\rangle$) with a change of the impact parameter of 0.2 Å when the H atoms occupy the fcc-adsorption sites at the H height of 1.2 Å. The incident beams are deflected by the shadowing effects of the Pt atoms on the first layer, then recoiling the H atoms as the DR-H (direct recoiled hydrogen) to 45° with respect to the incident beam. The small angle changes of DR-H as a function of the impact parameter for 0.2 Å show that the cross-sections for DR-H have the maximum values at the H height of 1.2 Å at these angles, respectively. The LERS experimental data and the calculations of $\alpha_{c,sh}$ and $\alpha_{c,bl}$ thus result in the final structural model shown in Fig. 11. In this model, the H atoms occupy fcc-adsorption sites on the Pt(111) terraces at a height of approximately 1.2 Å above the center of mass position of the first layer of Pt atoms, corresponding to a bond length of 1.9 ± 0.1 Å. This bond length is almost in good agreement with the reported value of 1.76 Å as found by an EELS study [6].

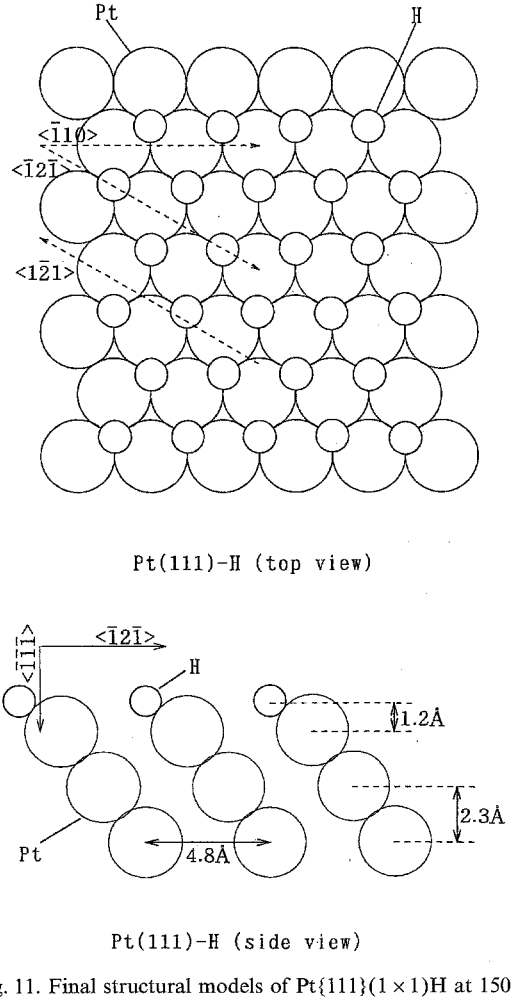


Fig. 11. Final structural models of Pt{111}(1 × 1)H at 150 K.

4. Conclusion

We have shown that LERS can be used to detect hydrogen on a Pt(111) surface at 150 K, and that LERS provides a direct probe for locating the positions of adsorbed hydrogen on a surface. This method complements previously used methods for observing the behavior of light adsorbates on solid surfaces. It was found: (1) the initial sticking coefficient of a H atom is $s_0(\theta)\approx0.1$; (2) from the isothermal annealing, the activation energy of H atom is evaluated to be 0.3 eV; and (3) the H atoms occupy fcc-adsorption sites on the Pt(111) terraces at a height of approximately 1.2 Å above the first layer Pt atoms.

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